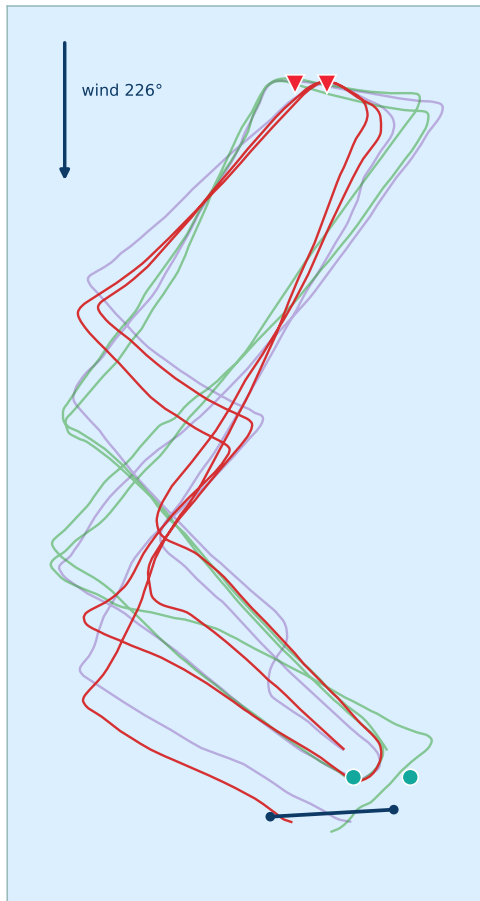


MOTH — Race 9



Lago di Garda, Malcesine · 5 Jul 2026 · Start 14:37 (local)

Race course — all boats



Fleet

CRO 5087

AUS 5250 *

ESP 56

Race winner

- CRO 5087 finished first in this race.
- 2nd: AUS 5250 * 3rd: ESP 56
- Afternoon breeze; W-L-W-L course.
- Wind, marks and course axis estimated from GPS (no wind sensor in logs).

Instrument note (*)

- AUS 5250: speed sensor mounted rotated. Heel & heel-stability are rebuilt from the boat's gravity + course — an accurate estimate, not exact to the degree.

Positions at the marks — gains (green) / losses (red)

Finish	Boat	M1	M2	M3	Fin	Δ M1→M2	Δ M2→M3	Δ M3→Fin	Net
1	CRO 5087	1	1	1	1	+0	+0	+0	+0
2	AUS 5250 *	2	2	2	2	+0	+0	+0	+0
3	ESP 56	3	3	3	3	+0	+0	+0	+0

M1 = windward 1 · M2 = leeward 1 · M3 = windward 2 · Fin = finish. Positive Δ (green) = places gained on that leg.



Fin	Boat	Line pos	Bias loss (m)	DTL gun	DTL -10s	SOG gun	SOG -10s	TWA gun	SOG 1'	Heel stab 1'	VMG 1'
1	CRO 5087	3-2	5	14	41	18.7	14.1	73	17.3	9.2	8.9
2	AUS 5250 *	2-2	15	39	73	22.6	15.4	70	18.9	19.0	10.5
3	ESP 56	2-1	20	20	44	21.6	20.1	81	16.6	6.4	7.6

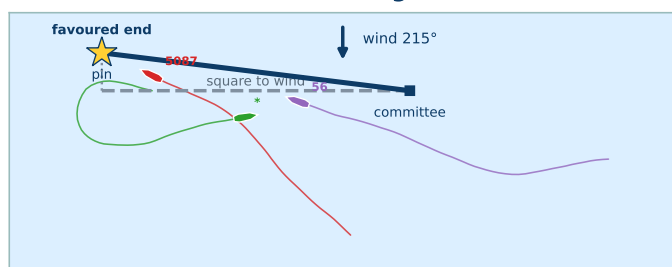
How to read the start

- Line pos: the line is split into 3 thirds, each third into 3 — read third-subpart from the committee end (1-1) to the pin end (3-3).
- Bias loss (m): metres given away at the gun from line position alone vs the favoured (pin) end, from the TWD bias.
- DTL = distance to line (m, negative = over early). 1' = first minute after the gun; heel stability = avg heel change per 10s (lower = steadier).

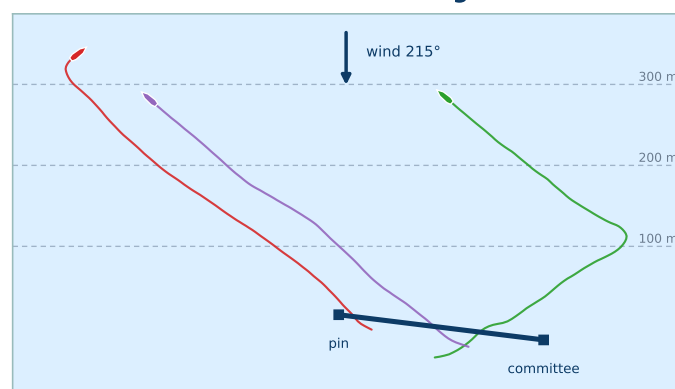
Line bias at the start (vs TWD)

- Start line 255 m, bearing 312°.
- Wind at start ≈ 215°, estimated from the first beat.
- Line bias vs TWD: 7.0° — Pin end favoured.
- Advantage ≈ 31 m (≈ 6.2 boat-lengths) for a boat starting at the pin end.

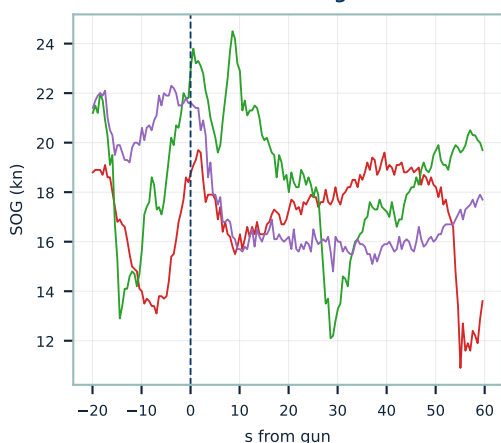
Position at gun



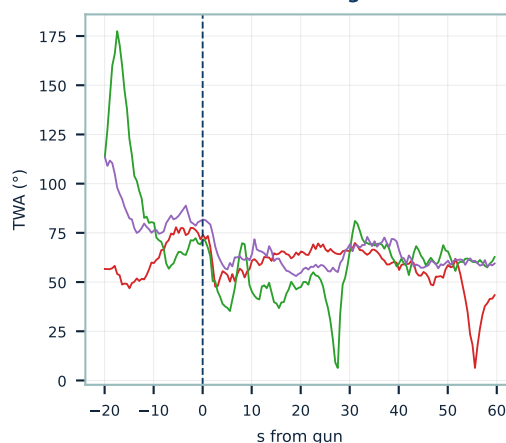
Position 1 min after gun



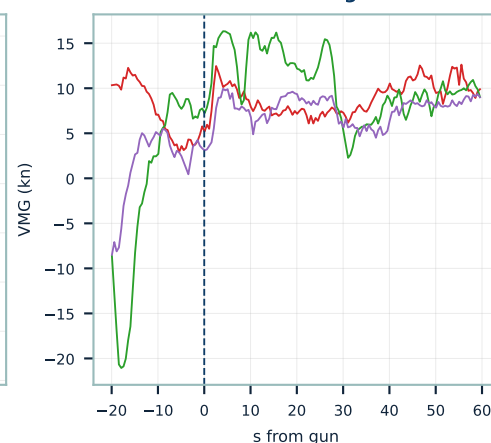
SOG around gun



TWA around gun



VMG around gun

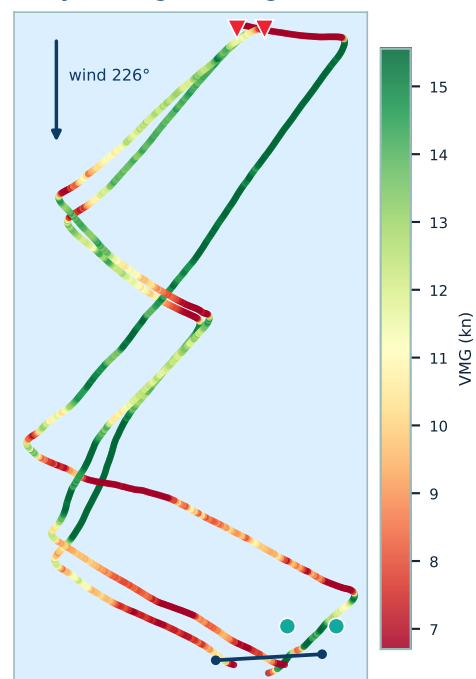




Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087 — AUS 5250* — ESP 56

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
1	CRO 5087	12.17	46	17.97	20.90	21.1	11.9	+0.3	+1.4	2251	244	3
2	AUS 5250*	11.05	52	18.93	22.61	22.4	10.8	-0.6	-1.7	2656	273	3
3	ESP 56	10.77	50	17.11	21.50	22.2	10.6	-2.3	+0.4	2414	275	3

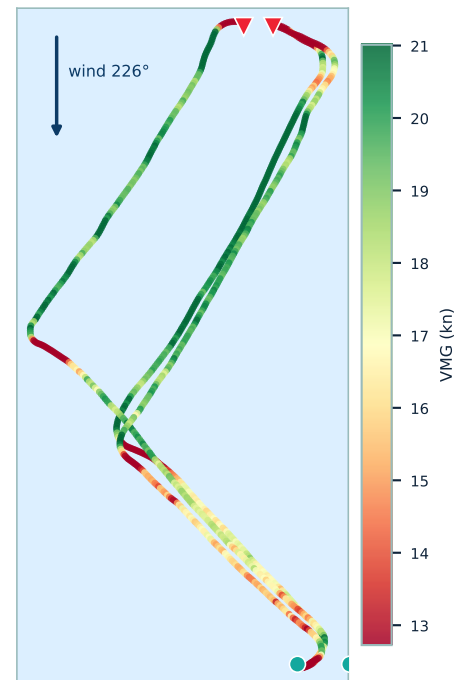
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087 — AUS 5250* — ESP 56

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	CRO 5087	17.44	142	22.67	25.00	13.4	6.9	+1.2	+1.0	1869	160	3
2	AUS 5250*	17.41	139	23.40	26.06	13.8	7.1	+0.9	-1.2	1937	160	2
3	ESP 56	16.77	142	22.01	24.34	9.6	8.5	+1.0	+0.3	1877	166	3

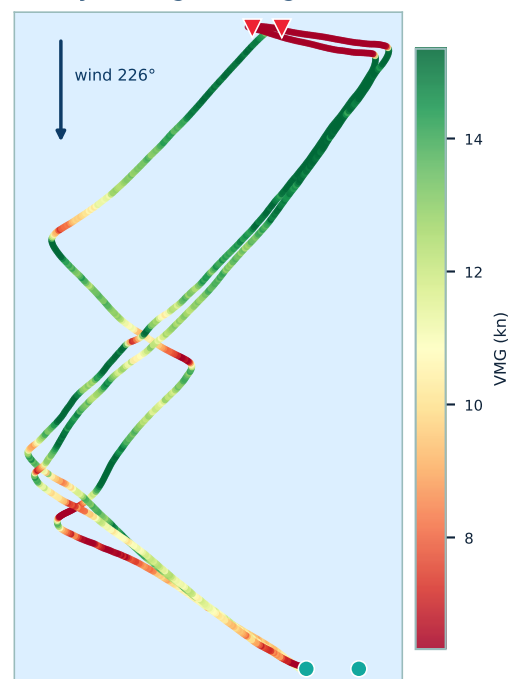
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087 — AUS 5250* — ESP 56

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Tacks
2	AUS 5250*	12.10	50	19.17	21.10	22.0	12.6	-0.7	-2.0	2278	231	2
1	CRO 5087	11.19	50	17.56	21.40	21.7	9.5	+0.6	+1.3	2260	250	3
3	ESP 56	11.04	50	17.75	19.50	22.7	8.2	-1.0	+1.3	2308	252	2

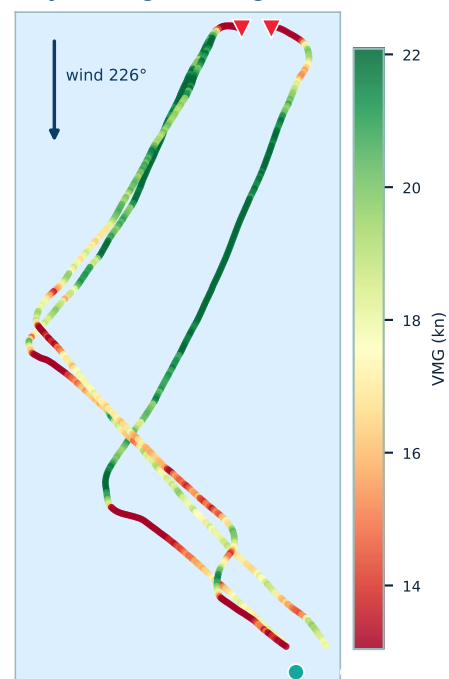
Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Tracks by boat



Tracks by VMG (green=high, red=low)



— CRO 5087 — AUS 5250* — ESP 56

Finish	Boat	Avg VMG	Avg TWA	Avg SOG	Max SOG	Heel_norm	Heel_stab	Heel Δ	Avg Pitch	Dist (m)	Time (s)	# Gybes
1	CRO 5087	18.41	145	23.22	26.62	12.6	6.9	+1.2	+1.1	1728	145	2
2	AUS 5250*	17.84	140	23.70	25.70	14.2	7.7	+5.1	-1.3	1821	149	1
3	ESP 56	16.60	141	21.90	24.10	8.4	7.7	+5.0	+0.5	1813	161	2

Rows ordered by Avg VMG (best first); the Finish column keeps each boats race finish position. Heel_norm = mean |heel|; Heel_stab = avg heel change per 10s (lower = steadier); Heel Δ = + more heel to starboard / - more to port; Avg Pitch + = bow up; Dist = distance sailed; Time = leg duration.



Finish	Boat	# Tacks	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
3	ESP 56	5	17.8	13.9	12.0	5.8	+7	122	21	19	21.6	25.4
2	AUS 5250 *	5	18.9	14.9	12.2	6.7	+14	135	20	20	22.5	33.5
1	CRO 5087	6	17.6	13.1	10.8	6.8	+6	103	19	21	25.7	24.8

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

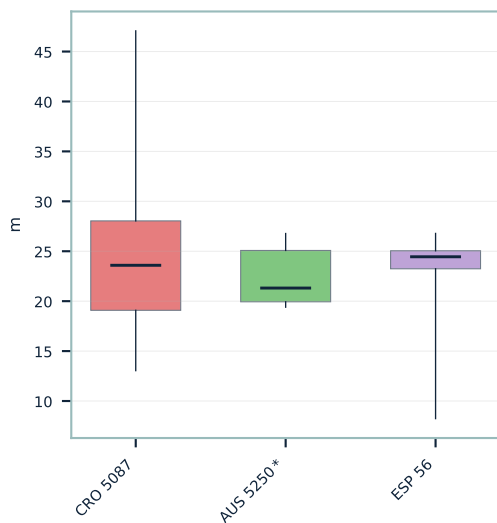
Tacks analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

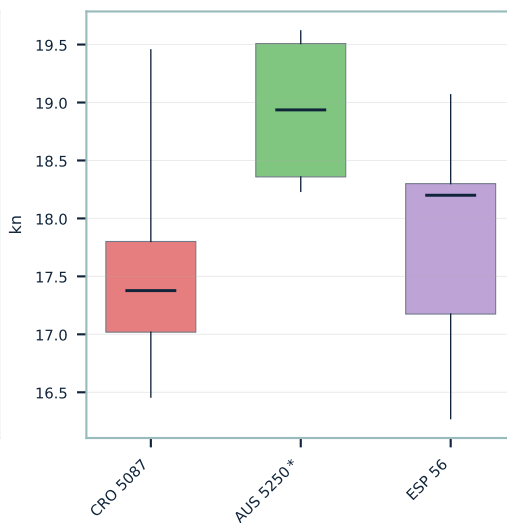
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

Dist loss (m, lower=better)



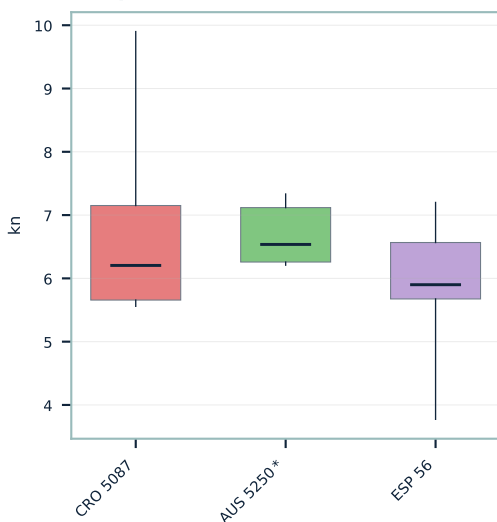
Entry speed (higher = better)



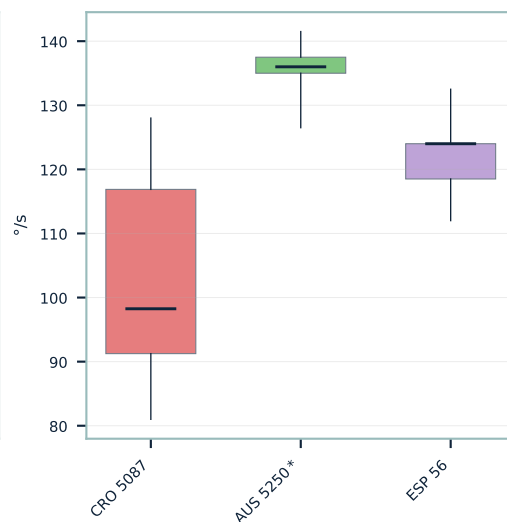
Min speed (higher = better)



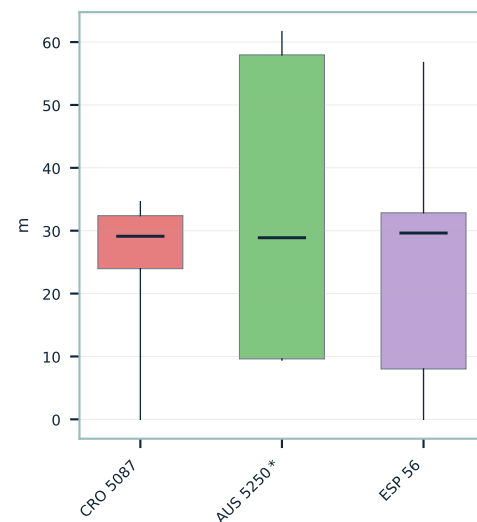
Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)





Finish	Boat	# Gybes	Entry	Exit	Min	SOG Δ	Angle Δ	Turn °/s	Heel in	Heel out	Dist loss (m)	VMG loss (m)
3	ESP 56	5	22.1	18.8	17.1	5.0	-2	77	12	8	17.8	16.8
1	CRO 5087	5	22.8	18.9	17.5	5.3	-5	64	12	9	21.4	24.0
2	AUS 5250 *	3	23.4	18.6	17.0	6.5	-12	98	12	9	27.5	39.0

Rows ordered by Dist loss (most efficient first); the Finish column keeps each boat's race finish position.

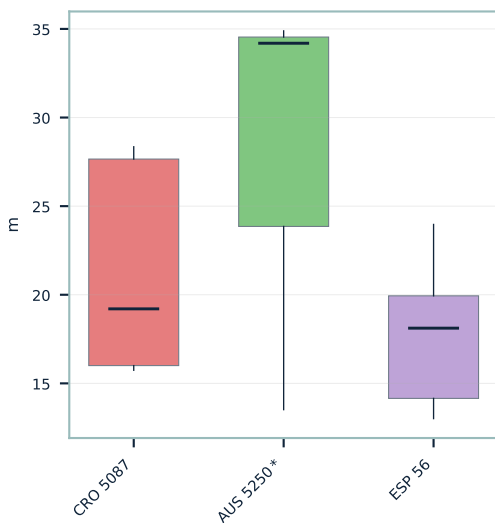
Gybes analysis — metric definitions

- Entry / Exit SOG = mean speed over 6 s before / 9 s after the manoeuvre. Min SOG = lowest speed in between.
- SOG Δ = Entry – Min (how much speed was bled off). Angle Δ = exit TWA – entry TWA (+ = exits wider).
- Rate of turn = peak heading change (°/s). Heel in/out = mean |heel|.
- VMG loss = made-good (toward the mark) deficit vs holding entry VMG. Dist loss = actual distance lost from the speed drop vs holding entry speed.

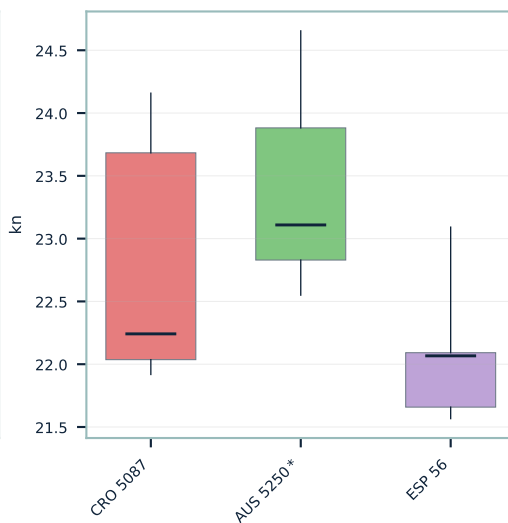
How to read the charts below

- Each box = one boat, ordered left→right by race finish. The plots span all of that boat's manoeuvres.
- Whiskers = min & max; box = middle 50% (25th–75th pct); line in the box = median. A lower, tighter box = more consistent.
- VMG/Dist loss, Speed loss, Min-speed: lower is better. Entry & Min speed: higher is better.

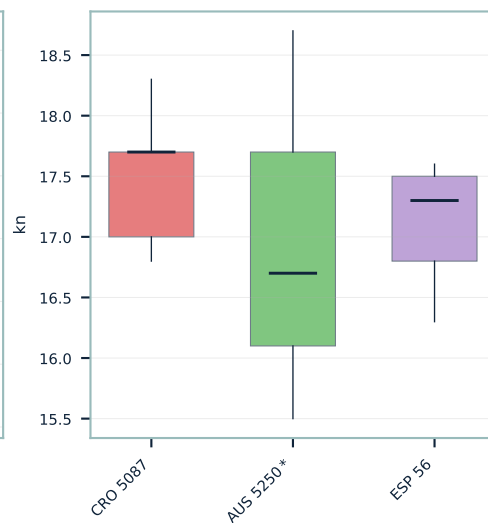
Dist loss (m, lower=better)



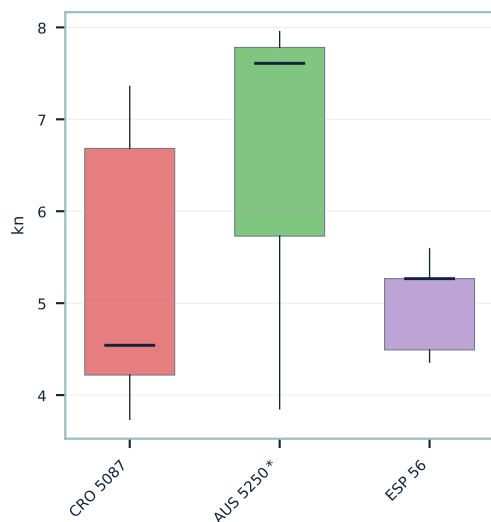
Entry speed (higher = better)



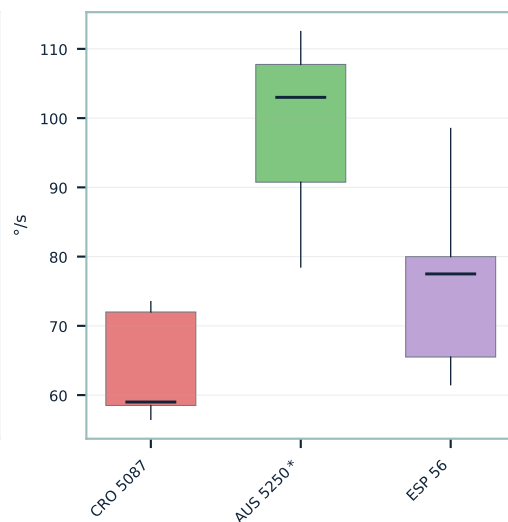
Min speed (higher = better)



Speed loss SOG Δ (lower = better)



Rate of turn



VMG loss (m, lower=better)

